

Meter-file fixes, and two explanations

9 October 2026, on v8. The scenario tests are merged into v8, so the golden-households test runs on every push. Fixes 1 to 4 are in, one commit each, and each one tightened the golden limits for the homes it put right; fixes 12 and 13 followed (the next section). Every number here comes from a run of the real app against homes whose true use and costs are known: all 71 scenarios on the build before each fix and after it.

- **All 71 scenarios: 29 passed before, 38 after fix 4, 36 after fixes 12 and 13** (see the next section for why). Fix 1 (a lone midnight reading counted as a day): nine-month files of all three homes pass. Fix 2 (each half hour once): a doubled file reads 0.0% where it read +99.5%, and a file the app cannot read 60% of now says so. Fix 4 (ask instead of overriding): "no panels" with sales in the file -34.2% to -1.8%, a file from a previous home +87.1% to +5.0%.
- **The accuracy figure now moves (fix 3).** It covers the error found in 44 of 62 accepted files (31 before). Four winter weeks of a heat pump home show $\pm 55\%$ where they showed $\pm 4\%$.
- **The golden test holds 20 homes** (14 before): the four user errors you named, D2a (must warn) and D6 (must say the file disagrees with what was typed). It now also holds the warnings, the questions asked and the accuracy figure.
- **B1 (question 4): the bill is negative, and the scoring compared two different years.** The app's bill for the tester-like home is negative in 2 of the 4 files. The percentages are large because the true bill is small: €332 a year. The "true" figure was the home's recorded year, which had 11 days without panels and six weeks of unpaid sales. The coming year with the panels up costs €275 on the same plan. The rest of the gap comes from what each file shows of the home: a year before the panels that was milder, and a summer or winter scaled up to a year.
- **The saving bias (question 5) has three causes, each measured.** For homes planning panels, two of the app's habits put together reproduce its figures to the kWh: its solar runs on winter time all summer, an hour early against the meter's clock, and its panels make 1.9% less than PVGIS. For battery homes the gap is all on the "without panels" side, which is worked back from the file. That home comes out 4% too small, and 41% to 55% of its use lands in the night hours, against 28.5% in truth. A night-rate plan then looks €137 to €300 a year cheaper than it is.
- **Real homes have a slot** (tests/fixtures/meter-scenarios/real/). Your ESB file with the Sigenergy export, and the tester's file, become scenarios with true figures. They stay out of git. A self-test on a made-up home recovers the stored truth exactly.

Fixes 12 and 13 (approved after the first version of this report)

Fix 12 puts the panels on the clock the meter keeps. Fix 13 stops a meter file with panels in it setting the hours of the home without them; its days before the panels do, or the heating profile does. All 71 scenarios were run on each. The golden test now also holds the solar saving, at 3%.

ID	Scenario	Before 12	Fix 12	Fix 13	Now	Saving limit
A1-gas	Calendar year 2025	saving -3.9% payback 7.1 v 6.82 bill -0.9%	saving -2.0% payback 7.0 v 6.82 bill -0.9%	saving -2.0% payback 7.0 v 6.82 bill -0.9%	pass	3%
A1-heatpump	Calendar year 2025	saving -5.1% payback 6.1 v 5.78 bill -4.6%	saving -2.9% payback 6.0 v 5.78 bill -4.6%	saving -2.9% payback 6.0 v 5.78 bill -4.6%	FAIL: consumption	not in the set
A1-ev	Calendar year 2025	saving -4.0% payback 6.7 v 6.43 bill -0.1%	saving -1.4% payback 6.5 v 6.43 bill -0.1%	saving -1.4% payback 6.5 v 6.43 bill -0.1%	pass	not in the set
A2-heatpump	Twelve months, March 2025 to February 2026	saving -4.7% payback 6.1 v 5.78 bill +0.6%	saving -2.5% payback 5.9 v 5.78 bill +0.6%	saving -2.5% payback 5.9 v 5.78 bill +0.6%	pass	3%
A7-ev	Two years, October 2024 to October 2026	saving -4.5% payback 6.7 v 6.43 bill 0.0%	saving -1.8% payback 6.5 v 6.43 bill 0.0%	saving -1.8% payback 6.5 v 6.43 bill 0.0%	pass	3%
A3a-gas	Nine months, June 2025 to February 2026 (no spring)	saving -1.2% payback 6.9 v 6.82 bill +1.2%	saving +1.6% payback 6.7 v 6.82 bill +1.2%	saving +1.6% payback 6.7 v 6.82 bill +1.2%	pass	3%
B2-gas_solar	A year after the panels, exports recorded	saving -2.3% payback 6.9 v 6.77 bill 0.0%	saving -2.3% payback 6.9 v 6.77 bill 0.0%	saving -2.3% payback 6.9 v 6.77 bill 0.0%	pass	3%
B2-hp_solar_batt	A year after the panels, exports recorded	saving -14.3% payback 7.2 v 6.2 bill +0.1%	saving -14.3% payback 7.2 v 6.2 bill +0.1%	saving -15.3% payback 7.3 v 6.2 bill +0.1%	FAIL: payback	16%

ID	Scenario	Before 12	Fix 12	Fix 13	Now	Saving limit
B5-hp_solar_gridfill	Battery filled from the grid at night in winter	saving -22.4% payback 7.7 v 5.97 bill +0.2%	saving -22.4% payback 7.7 v 5.97 bill +0.2%	saving -13.8% payback 6.9 v 5.97 bill +0.2%	FAIL: payback	15%
B3-friend	Panels from 20 Oct 2025, exports recorded only from 1 Dec 2025 (the tester's case)	saving -3.0% payback 5.9 v 5.72 bill +0.5%	saving -3.0% payback 5.9 v 5.72 bill +0.5%	saving -4.0% payback 6.0 v 5.72 bill +0.5%	pass	5% (loosened, reason recorded)
B4a-gas_solar	Before and after the panels (12 May 2025), exports recorded; user confirms the date	saving -2.5% payback 6.9 v 6.77 bill -1.9%	saving -0.9% payback 6.8 v 6.77 bill -4.4%	saving -0.9% payback 6.8 v 6.77 bill -4.4%	pass	3%
B4c-gas_solar	Before and after the panels, no export rows; user confirms the date shown	saving -3.5% payback 7.0 v 6.77 bill -0.7%	saving -0.9% payback 6.8 v 6.77 bill -4.9%	saving -0.6% payback 6.8 v 6.77 bill -5.3%	FAIL: bill	not in the set
B4d-hp_solar_batt	Before and after a battery system (3 Mar 2025), exports recorded; user confirms the date shown	saving +12.3% payback 5.5 v 6.2 bill -6.9%	saving +11.6% payback 5.6 v 6.2 bill -7.9%	saving +11.9% payback 5.5 v 6.2 bill -7.8%	FAIL: consumption, bill, payback	not in the set
C3-moved	Moved house: the file is the old (heat pump) home; the new home is gas-heated	saving +7.1% payback 6.4 v 6.82 bill +4.0%	saving +9.2% payback 6.2 v 6.82 bill +4.0%	saving +9.2% payback 6.2 v 6.82 bill +4.0%	FAIL: payback	10.5% (loosened, reason recorded)

- **Fix 12 did what was measured.** On the twelve full-year files the saving moves from -5.1%..-3.7% to -2.9%..-1.4%, and every payback shortens by 0.1 to 0.2 years toward the truth. What is left is mostly the panels' 1.9% shortfall against PVGIS (proposal 15). Shorter files, B4 and the D files move the same way. One new fail: C3, the person who moved. They type the typical 4,200 kWh for a 4,000 kWh home, priced on the standard profile, and its payback now reads 6.2 against 6.82. The early-hour bias had been hiding that; its limit is loosened in golden.json with that reason.
- **Fix 13 helped the battery filled from the grid, and not the others.** B5's saving went from -22.4% to -13.8%. B2-hp_solar_batt went from -14.3% to -15.3%, and the tester-like home B3-friend from -3.0% to -4.0%. These homes have no days before their panels, so the heat pump profile now sets the hours. That profile assumes the water is heated from 2am to 5am, the app's default for every heat pump home: 42% of the day lands in the night hours (on 16 January, 4.0 kWh an hour from 2am to 5am), against 28.5% in the true home. That is no better than the 41.1% the battery's pattern gave. B3-friend's saving limit is loosened with that reason.

- **That assumption is most of what is left.** Run again with the hot water left out of the profile (a measurement, not a change): B2-hp_solar_batt -6.1%, B5 -4.9%, B3-friend +3.5%. What then remains for B2 and B5 is mostly the home read about 4% small (proposal 14). The made-up homes do not heat water at night, so leaving it out would only tune the app to them; whether a real heat pump does is something the person knows. Proposal 17 below asks them.
- **B4c tips over.** A gas home's file from before and after its panels, with no export rows. Its usage pattern now comes from its days before the panels, which is the home's real use. Its bill moved from -4.9% to -5.3%, just past 5%. The rest of its error is the panels' shortfall and the app working in hours where the meter has half hours.

Build	Commit	Pass	Fail	Turned away	Accuracy shown covers the error
Before 12	d08af36	38	24	9	44 of 62
Fix 12	e8339ca	37	25	9	44 of 62
Fix 13	c3354e4	36	26	9	44 of 62

#	Proposal (for your approval)	Impact	Effort	Scenarios
17	Ask a heat pump home when it heats its water (and when the heating runs) wherever its meter file cannot show it: a file with panels in all of it. Today the app assumes 2am to 5am.	High for battery homes: saving -15.3% to -6.1% for B2-hp_solar_batt if the answer is 'during the day'	Small	B2-hp_solar_batt, B3-friend, B5

The golden households, before and after

Each cell is the error against the truth: use / bill (payback in years against the true one, where the app shows one), on fixes 1 to 4. Limits are use / bill / plan / payback (and, from fix 12, the solar saving); "acc" is the rule on the accuracy figure (covers the error, or may not fall below a figure). A limit only ever tightens. The exceptions are recorded in golden.json with their reasons, each a correct fix that removed an error which had been hiding another: A5b-heatpump, B3-friend, C3-moved.

Scenario	Limits before	Before	Fix 1	Fix 2	Fix 3	Fix 4	Limits now
A1-gas Calendar year 2025	5% / 5% / €25 / 0.5 y acc covers	-1.5% / -0.9% payback 7.1 v 6.82	-1.2% / -0.9% payback 7.1 v 6.82	-1.2% / -0.9% payback 7.1 v 6.82	-1.2% / -0.9% payback 7.1 v 6.82	-1.2% / -0.9% payback 7.1 v 6.82	5% / 5% / €25 / 0.5 y saving 3% acc covers asks: filehome

Scenario	Limits before	Before	Fix 1	Fix 2	Fix 3	Fix 4	Limits now
A2-heatpump Twelve months, March 2025 to February 2026	5% / 5% / €25 / 0.5 y acc covers	+0.5% / +0.6% payback 6.1 v 5.78	+0.8% / +0.6% payback 6.1 v 5.78	+0.8% / +0.6% payback 6.1 v 5.78	+0.8% / +0.6% payback 6.1 v 5.78	+0.8% / +0.6% payback 6.1 v 5.78	5% / 5% / €25 / 0.5 y saving 3% acc covers asks: filehome
A7-ev Two years, October 2024 to October 2026	5% / 5% / €25 / 0.5 y acc covers	+0.6% / 0.0% payback 6.7 v 6.43	+0.7% / 0.0% payback 6.7 v 6.43	+0.7% / 0.0% payback 6.7 v 6.43	+0.7% / 0.0% payback 6.7 v 6.43	+0.7% / 0.0% payback 6.7 v 6.43	5% / 5% / €25 / 0.5 y saving 3% acc covers asks: filehome
A3a-gas Nine months, June 2025 to February 2026 (no spring)	17.5% / 14% / €25 / 0.5 y acc ≥ ±4%	-15.9% / -12.8% payback 7.1 v 6.82	+1.5% / +1.2% payback 6.9 v 6.82	+1.5% / +1.2% payback 6.9 v 6.82	+1.5% / +1.2% payback 6.9 v 6.82	+1.5% / +1.2% payback 6.9 v 6.82	5% / 5% / €25 / 0.5 y saving 3% acc covers asks: filehome
A5b-heatpump Four winter weeks, January 2026	78% / 67.5% / €25 / 0.8 y acc ≥ ±4%	+73.6% / +63.6% payback 5.2 v 5.78	+79.9% / +69.1% payback 5.2 v 5.78	+79.9% / +69.1% payback 5.2 v 5.78	+79.9% / +69.1% payback 5.2 v 5.78	+79.9% / +69.1% payback 5.2 v 5.78	84.5% / 73.5% / €25 / 0.8 y saving 15% (loosened, reason recorded) acc ≥ ±55% asks: filehome
B2-gas_solar A year after the panels, exports recorded	5% / 5% / €25 / 0.5 y acc covers	-1.8% / 0.0% payback 6.9 v 6.77	-1.8% / 0.0% payback 6.9 v 6.77	-1.8% / 0.0% payback 6.9 v 6.77	-1.8% / 0.0% payback 6.9 v 6.77	-1.8% / 0.0% payback 6.9 v 6.77	5% / 5% / €25 / 0.5 y saving 3% acc covers asks: filehome
B2-hp_solar_batt A year after the panels, exports recorded	5% / 5% / €25 / 1.2 y acc ≥ ±4%	-4.4% / +0.1% payback 7.2 v 6.2	-4.4% / +0.1% payback 7.2 v 6.2	-4.4% / +0.1% payback 7.2 v 6.2	-4.4% / +0.1% payback 7.2 v 6.2	-4.4% / +0.1% payback 7.2 v 6.2	5% / 5% / €25 / 1.2 y saving 16% acc ≥ ±4% asks: filehome
B3-friend Panels from 20 Oct 2025, exports recorded only from 1 Dec 2025 (the tester's case)	5% / 5% / €25 / 0.5 y acc covers asks: filewhen	+3.7% / +0.5% payback 5.9 v 5.72	+3.7% / +0.5% payback 5.9 v 5.72	+3.7% / +0.5% payback 5.9 v 5.72	+3.7% / +0.5% payback 5.9 v 5.72	+3.7% / +0.5% payback 5.9 v 5.72	5% / 5% / €25 / 0.5 y saving 5% (loosened, reason recorded) acc covers asks: filehome, filewhen
B4a-gas_solar Before and after the panels (12 May 2025), exports recorded; user confirms the date	5% / 5% / €25 / 0.5 y acc covers asks: filewhen	-2.9% / -1.9% payback 6.9 v 6.77	-2.9% / -1.9% payback 6.9 v 6.77	-2.9% / -1.9% payback 6.9 v 6.77	-2.9% / -1.9% payback 6.9 v 6.77	-2.9% / -1.9% payback 6.9 v 6.77	5% / 5% / €25 / 0.5 y saving 3% acc covers asks: filehome, filewhen
B5-hp_solar_gridfill Battery filled from the grid at night in winter	5% / 5% / €25 / 1.9 y acc covers	-3.7% / +0.2% payback 7.7 v 5.97	-3.7% / +0.2% payback 7.7 v 5.97	-3.7% / +0.2% payback 7.7 v 5.97	-3.7% / +0.2% payback 7.7 v 5.97	-3.7% / +0.2% payback 7.7 v 5.97	5% / 5% / €25 / 1.1 y saving 15% acc covers asks: filehome

Scenario	Limits before	Before	Fix 1	Fix 2	Fix 3	Fix 4	Limits now
C1-gas_ev_now EV bought on 1 Apr 2026, halfway through the file	19% / 11.5% / €60 / 0.5 y acc ≥ ±4%	-17.2% / -10.1% payback 6.5 v 6.39	-16.9% / -10.1% payback 6.5 v 6.39	-16.9% / -10.1% payback 6.5 v 6.39	-16.9% / -10.1% payback 6.5 v 6.39	-16.9% / -10.1% payback 6.5 v 6.39	18.5% / 11.5% / €60 / 0.5 y saving 3% acc ≥ ±4% asks: filehome
D3b-gas Every row twice	105% / 85% / €25 / 0.5 y acc ≥ ±4%	+99.5% / +80.3% payback 6.6 v 6.82	+100.0% / +80.3% payback 6.6 v 6.82	0.0% / 0.0% payback 7.1 v 6.82	0.0% / 0.0% payback 7.1 v 6.82	0.0% / 0.0% payback 7.1 v 6.82	5% / 5% / €25 / 0.5 y saving 3% acc covers warns: dupes asks: filehome
D3c-gas Two overlapping files uploaded one after the other	5% / 5% / €25 / 0.5 y acc covers	-0.8% / 0.0% payback 7.1 v 6.82	-0.1% / 0.0% payback 7.1 v 6.82	-0.1% / 0.0% payback 7.1 v 6.82	-0.1% / 0.0% payback 7.1 v 6.82	-0.1% / 0.0% payback 7.1 v 6.82	5% / 5% / €25 / 0.5 y saving 3% acc covers asks: filehome
D5e-gas A PDF bill renamed .csv	turned away	turned away	turned away	turned away	turned away	turned away	turned away
B3-gas_solar A year after the panels, no export rows	36.5% / 88% / €25 / no payback yet acc ≥ ±4% asks: filewhen	-34.2% / +83.0% no payback	-34.0% / +83.0% no payback	-34.0% / +83.0% no payback	-34.0% / +83.0% no payback	-34.0% / +83.0% no payback	36.5% / 88% / €25 / no payback yet acc ≥ ±4% asks: filehome, filewhen
B6-gas_solar User says no solar; the file shows exports	36.5% / 5% / €25 / no payback yet acc ≥ ±7%	-34.2% / 0.0% no payback	-34.0% / 0.0% no payback	-34.0% / 0.0% no payback	-34.0% / 0.0% no payback	-1.8% / 0.0% payback 6.9 v 6.77	5% / 5% / €25 / 0.5 y saving 3% acc covers asks: fileexp, filehome
B8-gas_solar User says 14 panels; the home has 9	53% / 5% / €25 / 3 y acc ≥ ±4%	+49.9% / 0.0% payback 4.1 v 6.77	+49.9% / 0.0% payback 4.1 v 6.77	+49.9% / 0.0% payback 4.1 v 6.77	+49.9% / 0.0% payback 4.1 v 6.77	+49.9% / 0.0% payback 4.1 v 6.77	53% / 5% / €25 / 3 y saving 70% acc ≥ ±4% asks: filehome
C3-moved Moved house: the file is the old (heat pump) home; the new home is gas-heated	92% / 75% / €25 / 0.9 y acc ≥ ±4%	+87.1% / +70.5% payback 6.1 v 6.82	+87.5% / +70.5% payback 6.1 v 6.82	+87.5% / +70.5% payback 6.1 v 6.82	+87.5% / +70.5% payback 6.1 v 6.82	+5.0% / +4.0% payback 6.4 v 6.82	5% / 5% / €25 / 0.8 y saving 10.5% (loosened, reason recorded) acc covers asks: filehome
D2a-gas Edited in Excel: US dates (MM/DD)	not in the set	-1.5% / -1.2% payback 6.9 v 6.82 60.5% unread, no warning	-1.2% / -0.9% payback 6.9 v 6.82 60.5% unread, no warning	-1.2% / -0.9% payback 6.9 v 6.82 60.5% unread, warned	-1.2% / -0.9% payback 6.9 v 6.82 60.5% unread, warned	-1.2% / -0.9% payback 6.9 v 6.82 60.5% unread, warned	5% / 5% / €25 / 0.5 y saving 3% acc covers warns: unread asks: filehome

Scenario	Limits before	Before	Fix 1	Fix 2	Fix 3	Fix 4	Limits now
D6-gas Typed 5,600 kWh in setup, then uploaded a file showing 4,000	not in the set	-0.2% / 0.0% payback 7.1 v 6.82	0.0% / 0.0% payback 7.1 v 6.82	0.0% / 0.0% payback 7.1 v 6.82	0.0% / 0.0% payback 7.1 v 6.82	0.0% / 0.0% payback 7.1 v 6.82	5% / 5% / €25 / 0.5 y saving 3% acc covers warns: typed asks: filehome, typed

All 71 scenarios, fix by fix

Build	Commit	Pass	Fail	Turned away	Accuracy shown covers the error
Before	90c7db3	29	33	9	31 of 62
Fix 1	8f633d6	33	29	9	35 of 62
Fix 2	93b4f6d	36	26	9	37 of 62
Fix 3	edba4a1	36	26	9	42 of 62
Fix 4	a9a2db6	38	24	9	44 of 62
Now (with the card fix)	708262f	38	24	9	44 of 62

Every scenario whose figures changed, before and now. Use / bill / payback; the accuracy shown; the result.

ID	Scenario	Before	Now
A1-gas	Calendar year 2025	-1.5% / -0.9% payback 7.1 v 6.82 ±4% pass	-1.2% / -0.9% payback 7.1 v 6.82 ±4% pass
A1-heatpump	Calendar year 2025	-5.5% / -4.6% payback 6.1 v 5.78 ±4% FAIL: consumption	-5.2% / -4.6% payback 6.1 v 5.78 ±4% FAIL: consumption
A1-ev	Calendar year 2025	-0.8% / -0.1% payback 6.7 v 6.43 ±4% pass	-0.5% / -0.1% payback 6.7 v 6.43 ±4% pass
A2-gas	Twelve months, March 2025 to February 2026	-0.9% / -0.5% payback 7.1 v 6.82 ±4% pass	-0.7% / -0.5% payback 7.1 v 6.82 ±4% pass
A2-heatpump	Twelve months, March 2025 to February 2026	+0.5% / +0.6% payback 6.1 v 5.78 ±4% pass	+0.8% / +0.6% payback 6.1 v 5.78 ±4% pass

ID	Scenario	Before		Now	
A2-ev	Twelve months, March 2025 to February 2026	+1.8% / +0.7% payback 6.7 v 6.43 ±4%	pass	+2.1% / +0.7% payback 6.7 v 6.43 ±4%	pass
A3a-gas	Nine months, June 2025 to February 2026 (no spring)	-15.9% / -12.8% payback 7.1 v 6.82 ±4%	FAIL: consumption, bill	+1.5% / +1.2% payback 6.9 v 6.82 ±6%	pass
A3a-heatpump	Nine months, June 2025 to February 2026 (no spring)	-17.8% / -15.8% payback 6.3 v 5.78 ±4%	FAIL: consumption, bill	-1.1% / -1.0% payback 6.0 v 5.78 ±15%	pass
A3a-ev	Nine months, June 2025 to February 2026 (no spring)	-13.3% / -11.5% payback 6.7 v 6.43 ±4%	FAIL: consumption, bill, plan	+4.6% / +6.8% payback 6.4 v 6.43 ±6%	pass
A3b-gas	Nine months, September 2025 to May 2026 (no summer)	-0.9% / -0.7% payback 6.9 v 6.82 ±4%	pass	-0.3% / -0.2% payback 6.9 v 6.82 ±6%	pass
A3b-heatpump	Nine months, September 2025 to May 2026 (no summer)	+8.1% / +7.2% payback 5.8 v 5.78 ±4%	pass	+8.6% / +7.6% payback 5.8 v 5.78 ±15%	pass
A3b-ev	Nine months, September 2025 to May 2026 (no summer)	-2.6% / -1.8% payback 6.5 v 6.43 ±4%	FAIL: plan	-2.0% / -1.3% payback 6.5 v 6.43 ±6%	FAIL: plan
A4-gas	This year so far, 1 January to 8 October 2026	0.0% / 0.0% payback 6.9 v 6.82 ±4%	pass	+0.6% / +0.5% payback 6.9 v 6.82 ±5%	pass
A4-heatpump	This year so far, 1 January to 8 October 2026	-3.3% / -2.9% payback 6.0 v 5.78 ±4%	pass	-2.9% / -2.6% payback 6.0 v 5.78 ±15%	pass
A4-ev	This year so far, 1 January to 8 October 2026	+0.6% / +0.8% payback 6.5 v 6.43 ±4%	FAIL: plan	+1.1% / +1.2% payback 6.5 v 6.43 ±5%	FAIL: plan
A5a-gas	Four summer weeks, July 2026	-7.0% / -5.6% payback 7.1 v 6.82 ±4%	pass	-3.7% / -2.9% payback 7.1 v 6.82 ±15%	pass
A5a-heatpump	Four summer weeks, July 2026	-45.2% / -40.0% payback 6.7 v 5.78 ±4%	FAIL: consumption, bill	-43.3% / -38.3% payback 6.7 v 5.78 ±55%	FAIL: consumption, bill
A5a-ev	Four summer weeks, July 2026	-9.1% / -7.3% payback 6.5 v 6.43 ±4%	FAIL: plan	-5.9% / -4.4% payback 6.5 v 6.43 ±15%	FAIL: plan

ID	Scenario	Before		Now	
A5b-gas	Four winter weeks, January 2026	0.0% / 0.0% payback 7.0 v 6.82 ±4%	pass	+3.5% / +2.9% payback 7.0 v 6.82 ±15%	pass
A5b-heatpump	Four winter weeks, January 2026	+73.6% / +63.6% payback 5.2 v 5.78 ±4%	FAIL: consumption, bill	+79.9% / +69.1% payback 5.2 v 5.78 ±55%	FAIL: consumption, bill
A5b-ev	Four winter weeks, January 2026	-2.1% / -1.4% payback 6.6 v 6.43 ±4%	FAIL: plan	+1.4% / +1.7% payback 6.5 v 6.43 ±15%	FAIL: plan
A5c-gas	Two summer weeks, August 2026	-8.4% / -6.8% payback 6.8 v 6.82 ±4%	pass	-1.8% / -1.5% payback 6.8 v 6.82 ±15%	pass
A5c-heatpump	Two summer weeks, August 2026	-41.5% / -36.8% payback 6.6 v 5.78 ±4%	FAIL: consumption, bill	-37.4% / -33.1% payback 6.5 v 5.78 ±58%	FAIL: consumption, bill
A5c-ev	Two summer weeks, August 2026	-12.2% / -9.1% payback 6.5 v 6.43 ±4%	FAIL: plan	-5.9% / -3.7% payback 6.4 v 6.43 ±15%	FAIL: plan
A6-gas	An old year, July 2024 to June 2025	-1.6% / -1.2% payback 7.1 v 6.82 ±4%	pass	-1.4% / -1.2% payback 7.1 v 6.82 ±4%	pass
A6-heatpump	An old year, July 2024 to June 2025	-4.2% / -3.7% payback 6.1 v 5.78 ±4%	pass	-4.0% / -3.7% payback 6.1 v 5.78 ±4%	pass
A6-ev	An old year, July 2024 to June 2025	-0.1% / -0.3% payback 6.8 v 6.43 ±4%	pass	+0.1% / -0.3% payback 6.8 v 6.43 ±4%	pass
A7-gas	Two years, October 2024 to October 2026	-0.8% / 0.0% payback 7.1 v 6.82 ±4%	pass	-0.7% / 0.0% payback 7.1 v 6.82 ±4%	pass
A7-heatpump	Two years, October 2024 to October 2026	-2.0% / +0.1% payback 6.1 v 5.78 ±4%	pass	-1.9% / +0.1% payback 6.1 v 5.78 ±4%	pass
A7-ev	Two years, October 2024 to October 2026	+0.6% / 0.0% payback 6.7 v 6.43 ±4%	pass	+0.7% / 0.0% payback 6.7 v 6.43 ±4%	pass
B1a-friend	Full year before the panels	-4.5% / -46.3% payback 5.1 v 5.72 ±4%	FAIL: bill, payback	-4.3% / -46.3% payback 5.1 v 5.72 ±4%	FAIL: bill, payback

ID	Scenario	Before		Now	
B1b-friend	Half year before the panels	-23.7% / -145.9% payback 5.4 v 5.72 ±4%	FAIL: consumption, bill	-23.3% / -144.0% payback 5.3 v 5.72 ±30%	FAIL: consumption, bill
B1c-friend	Summer before the panels	-55.6% / -220.1% payback 5.8 v 5.72 ±4%	FAIL: consumption, bill, plan	-34.4% / -156.7% payback 5.5 v 5.72 ±45%	FAIL: consumption, bill, plan
B1d-friend	Winter before the panels	-14.6% / -77.7% payback 5.4 v 5.72 ±4%	FAIL: consumption, bill	+27.8% / +71.6% payback 4.5 v 5.72 ±45%	FAIL: consumption, bill, payback
B3-gas_solar	A year after the panels, no export rows	-34.2% / +83.0% no payback ±4%	FAIL: consumption, bill, no payback shown	-34.0% / +83.0% no payback ±4%	FAIL: consumption, bill, no payback shown
B4c-gas_solar	Before and after the panels, no export rows; user confirms the date shown	-3.0% / -0.7% payback 7.0 v 6.77 ±4%	pass	-3.0% / -0.7% payback 7.0 v 6.77 ±7%	pass
B6-gas_solar	User says no solar; the file shows exports	-34.2% / 0.0% no payback ±7%	FAIL: consumption, no payback shown	-1.8% / 0.0% payback 6.9 v 6.77 ±4%	pass
B7a-gas	User says they have 9 panels; the file shows a home without them (they have none)	-0.2% / 0.0% no payback ±4%	FAIL: no payback shown	0.0% / 0.0% no payback ±4%	FAIL: no payback shown
B7b-gas	Same; user answers that the file is from before the panels	-0.2% / -58.3% payback 7.1 v 6.82 ±4%	FAIL: bill	0.0% / -58.3% payback 7.1 v 6.82 ±4%	FAIL: bill
C1-gas_ev_now	EV bought on 1 Apr 2026, halfway through the file	-17.2% / -10.1% payback 6.5 v 6.39 ±4%	FAIL: consumption, bill, plan	-16.9% / -10.1% payback 6.5 v 6.39 ±4%	FAIL: consumption, bill, plan
C2-gas_to_hp	Heat pump put in on 15 Jan 2026 (was gas)	-17.4% / -15.4% payback 6.3 v 5.79 ±4%	FAIL: consumption, bill	-17.2% / -15.4% payback 6.3 v 5.79 ±4%	FAIL: consumption, bill
C4-gas_holiday	Away for four weeks in August 2026	-5.2% / -4.0% payback 7.2 v 6.82 ±4%	FAIL: consumption	-5.0% / -4.0% payback 7.2 v 6.82 ±4%	pass
C3-moved	Moved house: the file is the old (heat pump) home; the new home is gas-heated	+87.1% / +70.5% payback 6.1 v 6.82 ±4%	FAIL: consumption, bill, payback	+5.0% / +4.0% payback 6.4 v 6.82 ±6%	pass
D1b-gas	Half-hourly file in kWh, not kW	-0.3% / 0.0% payback 7.1 v 6.82 ±4%	pass	0.0% / 0.0% payback 7.1 v 6.82 ±4%	pass

ID	Scenario	Before		Now	
D2a-gas	Edited in Excel: US dates (MM/DD)	-1.5% / -1.2% payback 6.9 v 6.82 60.5% unread, no warning ±4%	FAIL: lost readings, no warning	-1.2% / -0.9% payback 6.9 v 6.82 60.5% unread, warned ±17%	pass
D2d-gas	Saved by Excel with Irish short dates (d/m/yyyy h:mm)	-0.2% / 0.0% payback 7.1 v 6.82 ±4%	pass	0.0% / 0.0% payback 7.1 v 6.82 ±4%	pass
D3a-gas	Missing days: ten single days and a three-week gap	-0.8% / -2.6% payback 7.1 v 6.82 ±4%	pass	-0.6% / -2.6% payback 7.1 v 6.82 ±4%	pass
D3b-gas	Every row twice	+99.5% / +80.3% payback 6.6 v 6.82 ±4%	FAIL: consumption, bill	0.0% / 0.0% payback 7.1 v 6.82 ±4%	pass
D3c-gas	Two overlapping files uploaded one after the other	-0.8% / 0.0% payback 7.1 v 6.82 ±4%	pass	-0.1% / 0.0% payback 7.1 v 6.82 ±4%	pass
D3d-gas	Meter replaced in March: two serial numbers in one file	-0.2% / 0.0% payback 7.1 v 6.82 ±4%	pass	0.0% / 0.0% payback 7.1 v 6.82 ±4%	pass
D4-gas	Calendar 2024: 29 February and both clock changes	-1.2% / -1.0% payback 7.1 v 6.82 ±4%	pass	-0.9% / -1.0% payback 7.1 v 6.82 ±4%	pass
D5f-gas	A very large file (the same 33 months five times, about 14 MB)	+396.5% / +321.4% payback 5.9 v 6.82 ±4%	FAIL: consumption, bill, payback	-0.6% / 0.0% payback 7.1 v 6.82 ±4%	pass
D6-gas	Typed 5,600 kWh in setup, then uploaded a file showing 4,000	-0.2% / 0.0% payback 7.1 v 6.82 ±4%	pass	0.0% / 0.0% payback 7.1 v 6.82 ±4%	pass

What the person sees now

✓ Imported 17,520 readings

Jan-Feb: 697 kWh · Mar-Apr: 664 kWh · May-Jun: 588 kWh · Jul-Aug: 638 kWh · Sep-Oct: 693 kWh · Nov-Dec: 721 kWh

Total: 4,001 kWh/yr — anticipated full-year profile

Readings in kW (avg per 30-min interval) — converted ×0.5 to kWh

✓ A full year of readings.

Your file runs 9 Oct 2025 to 8 Oct 2026. No solar shows in it.

You told us 5,600 kWh a year. This file says 4,001. We'll use the file unless you'd rather keep yours.

Keep my 5,600 kWh

Is this file from the home you live in now?

Yes, use this data →

No, a home I lived in before

Typed 5,600 kWh in setup, then a file showing 4,001: said, with the choice to keep it (fix 4). Every upload now asks whether the file is from the home lived in now.

✓ Imported 6,912 readings

Jan-Feb: 623 kWh · Mar-Apr: 701 kWh · May-Jun: 628 kWh · Jul-Aug: 676 kWh · Sep-Oct: 671 kWh · Nov-Dec: 654 kWh

Total: 3,953 kWh/yr — anticipated full-year profile

Readings in kW (avg per 30-min interval) — converted ×0.5 to kWh

Partial year: 144 days across 6 of 6 billing periods. We filled the gaps from the rest. A full year would make the seasons more accurate.

We couldn't read 10,606 of your 17,518 readings. Their dates aren't written the way ESB writes them. Excel often changes them when a file is opened and saved. Download it again from ESB Networks and add it as it comes. Until then, these figures use the rest.

2 readings were repeats of others in the file. We count each half hour once.

Your file runs 10 Jan 2025 to 9 Dec 2026. No solar shows in it.

Is this file from the home you live in now?

Yes, use this data →

No, a home I lived in before

A file opened and saved in Excel with US dates: 60.5% of its readings unreadable, and the card says so (fix 2).

✓ Imported 17,520 readings

Jan-Feb: 697 kWh · Mar-Apr: 664 kWh · May-Jun: 588 kWh · Jul-Aug: 638 kWh · Sep-Oct: 693 kWh · Nov-Dec: 721 kWh

Total: 4,001 kWh/yr — anticipated full-year profile

Readings in kW (avg per 30-min interval) — converted ×0.5 to kWh

✓ A full year of readings.

17,520 readings were repeats of others in the file. We count each half hour once.

Your file runs 9 Oct 2025 to 8 Oct 2026. No solar shows in it.

Is this file from the home you live in now?

Yes, use this data →

No, a home I lived in before

Every row twice: counted once, and said (fix 2).

Step 9 of 13 · Solar

Your meter file shows power sold from 9 Oct 2025. Do you have solar panels?

A meter only records power sold when something at the home makes it.

Yes, I have panels

We'll ask about them next

No

We'll go by your answer

"No panels", with sales in the file from the first day: asked (fix 4).

Question 4: the B1 bills of -46% to -220%

B1 is the tester-like home (heat pump, 22 panels on two faces, a 9 kWh battery that stores only solar) with a file from before the panels. The app adds the stated panels and battery to the file and prices the year. The scored figure is the app's answer for a battery that stores only solar; its headline assumes night filling and is shown beside it.

ID	File	True use in the file	App's year	App's plan	App: bill	App: headline	True: recorded year	True: coming year	Now (after fix 1)
B1a-friend	Full year before the panels	6,221 kWh in 365 days	6,206 (-4.5%)	Electric Ireland Home Electric + SST Saver 16%	€178	€26	€332	€289	6,219, €178
B1b-friend	Half year before the panels	2,298 kWh in 183 days	4,960 (-23.7%)	Electric Ireland Home Electric + SST Saver 16%	-€152	-€247	€332	€289	4,985, -€146
B1c-friend	Summer before the panels	1,015 kWh in 92 days	2,884 (-55.6%)	Pinergy Lifestyle Family Time	-€693	-€710	€577	€633	4,264, -€327

ID	File	True use in the file	App's year	App's plan	App: bill	App: headline	True: recorded year	True: coming year	Now (after fix 1)
B1d-friend	Winter before the panels	2,119 kWh in 90 days	5,551 (-14.6%)	Electric Ireland Home Electric + SST Saver 16%	€74	-€64	€332	€289	8,304, €569

- **Yes, the bill is negative.** On the scored answer it is negative for 2 of the 4 files; on the headline, for 3. A 9.7 kWp system with a battery takes this home from €2,184 a year without panels to €275 to €332 with them, so a negative bill says the panels would earn more than the home spends.
- **The percentages are large because the true figure is small.** €100 on a €332 bill is 30%. In euro, the errors are €154, €484, €1,270 and €258.
- **The scoring did compare two different things.** The true bill was the home's recorded year (9 Oct 2025 to 8 Oct 2026): eleven days before the panels went up, then six weeks when ESB did not yet record sales, so they were not paid. The app works out the coming year, with the panels up and sales paid from the first day. That year costs €275 on Electric Ireland's SST plan with the reference battery, or €289 with a battery run as the app runs its own (92% round trip, 10% kept back, held through the night). The recorded year costs €332. Against the coming year, B1a's error is €111 (-38%), not -46%. The B3-friend file is that recorded year, priced from its own readings, so there the two agree: €333 against €332.
- **What is left is the file.** B1a's year before the panels used 6,221 kWh, and the app read 6,206. That year was milder than the coming one (6,500 kWh), and the app's panels on these two faces make +148 kWh against PVGIS. So it buys less and sells more. B1b to B1d scale a half year, a summer or a winter up to a year, as the partial files A3 to A5 do. A heat pump home's summer is a third of its winter, so the year is read low from a summer and high from a winter. Fix 1 changed these (B1d now reads high: it had been held down by the lone midnight reading). Fix 3 now shows ±30% to ±45% on them, where it showed ±4%.
- The headline's night filling is a separate matter. On these plans it moves the answer to Electric Ireland Night Boost, €118 a year dearer than the solar-only best: that is the "Ask how a battery charges also when a file is uploaded" item (fix 6). Holding the battery through the night, as the app does on a night-rate plan, costs this home only €3 a year (€275 against €278 with the reference battery), so it is not a cause.

So the scoring change for later: score a file from before the panels against the coming year with the panels up all year, as above, not against the recorded year. B1a would then read -38% (-€111) rather than -46.3%.

Question 5: where the saving bias comes from

The saving is the cheapest plan without the panels less the cheapest plan with them, for the coming year. For homes planning panels the app's figure was 3.7% to 5.0% low; for battery homes 14% to 22% low. Each part below was measured: on the app (its own simulation, read out of the running page) and on the true homes (the truth's simulation, with one of the app's habits switched on at a time).

Homes planning panels: two habits, both in the with-solar year

Home	True saving	Solar an hour early in summer	Panels 1.9% lower	Both	The app
gas (A7-gas)	€880 sold 2,556, used 1,357	€862 (-2.0%) used 1,285	€864 (-1.8%)	€847 bought 2,724, sold 2,565	€847 bought 2,723, sold 2,565
heat pump (A7-heatpump)	€1,037 sold 1,919, used 1,994	€1,015 (-2.1%) used 1,965	€1,020 (-1.6%)	€999 bought 5,553, sold 1,893	€985 bought 5,630, sold 1,965
EV (A7-ev)	€934 sold 2,644, used 1,270	€909 (-2.6%) used 1,195	€918 (-1.7%)	€893 bought 4,813, sold 2,654	€892 bought 4,816, sold 2,657

- **The app's solar runs on winter time all year.** On 21 June its 4 kWp system makes power from 04:00 and peaks at 12:00 to 13:00. The sun is highest at about 13:30 Irish summer time. The meter's readings, and every plan's hours, are on the clock. So from late March to late October the panels run an hour early against the home: less is used at home and more sold for less. On the true gas home this alone takes 72 kWh off what is used at home and 2.0% off the saving.
- **The app's panels make 1.9% less than PVGIS** for 4 kWp facing south at 35° in Cork (3,841 kWh against 3,914). On the tester-like home's two faces it is the other way, +1.9% (7,777 against 7,629). So it is the panel model's allowance for direction that is off, not its overall level.
- **The two together reproduce the app.** On the gas home: bought 2,724 kWh (app 2,723), sold 2,565 (app 2,565), saving €847 (app €847). The no-panels side is right: €1,453 against €1,453. The heat pump and EV homes come within €14, the rest being their use read slightly low.

Battery homes: the "without panels" home worked back from the file

Home	With panels: app / true	Home without panels: kWh, app / true	On a flat-rate plan: app / true	Night share 23:00-08:00: app / true	App's cheapest without panels	Saving: app / true
B2-hp_solar_batt	€847 / €846	7,174 / 7,500	€2,381 / €2,476 (–€95)	41.1% / 28.5%	Electric Ireland Home Electric + SST Saver 16%, €2,244: €137 under the flat plan	€1,397 / €1,630 (-14.3%)

Home	With panels: app / true	Home without panels: kWh, app / true	On a flat-rate plan: app / true	Night share 23:00-08:00: app / true	App's cheapest without panels	Saving: app / true
B5-hp_solar_gridfill	€785 / €784	7,233 / 7,500	€2,398 / €2,476 (–€78)	54.7% / 28.5%	Electric Ireland Home Electric + SST Saver 16%, €2,098: €300 under the flat plan	€1,313 / €1,692 (-22.4%)

- **With the panels, the app is exact:** these files are priced from their own readings.
- **The home without panels is too small:** 7,174 kWh against 7,500. It is worked back as what was bought, plus what the stated panels make in a typical year, less what was sold, less 8% of the solar used for battery losses. The app's 6 kWp makes 5,737 kWh in its typical year against 5,905 in this one. Its battery allowance takes off 260 kWh where the battery lost 93. On a flat-rate plan that is the €95 and €78 above.
- **And its hours are wrong.** They come from the usage pattern read off what the meter bought. A battery has already moved that buying into the night, and a battery filled from the grid in winter more so. So 41.1% and 54.7% of the no-panels home's use land between 23:00 and 08:00, against 28.5% in the true home. On 16 January it uses 0.7 kWh an hour through the day and 1.9 through the night. A night-rate plan then looks €137 and €300 cheaper than a flat one, where in truth it is €13 dearer.

What the explanations point to (for your approval)

Not implemented. Ranked as before, impact against effort, and continuing the earlier list.

#	Fix	Impact	Effort	Scenarios
12	Put the panels on the clock the meter keeps (summer time from late March to late October)	High: every solar saving and payback, +2.0% to +2.6%	Small	A, B
13	Work the battery home's no-panels hours out from its days before the panels, or the heating profile, not from what the meter bought	High for battery homes: saving -14% and -22% today	Small	B2-hp_solar_batt, B5, B3-friend
14	Battery losses when working use back: the battery's own losses in place of 8% of the solar used	Medium: use –4% on battery homes	Small	B2, B4d, B5
15	Check the panel model against PVGIS face by face (-1.9% facing south, +1.9% on south-west and north-east)	Medium	Medium	All solar
16	Score a file from before the panels against the coming year with the panels up all year	Scoring only	Small	B1

Fix 5, later: as you set it out

For a solar file with no export rows yet, export data will not be made up. The answers will include "Panels were up the whole time; sales aren't on the file yet". With it, the result is shown as an estimate: the home's use is worked back from the stated panels, the sales are what those panels would sell, and the accuracy figure is widened for both. The golden homes for it are already in place: B3-gas_solar today reads -34.0% use, +83.0% bill and no payback, and must show a payback once fixed.

Real homes: ready for your files

- Folder: `tests/fixtures/meter-scenarios/real/private/<id>/` (git ignores it). Put in it the ESB file (My Meter, Downloads, "30-minute readings in kW"), the Sigenenergy export for the same twelve months at its finest interval (CSV or Excel), and `home.json` from `home.example.json`: your setup answers and the system as installed.
- `add_home.py <folder>` replaces the MPRN and meter serial. It reads the Sigenenergy columns by their names and finds out whether its rows are stamped at the start or end of each interval, by lining its sales up with the meter file's. It prints both files' bought and sold totals side by side, then writes the truth: the bills on every plan from the meter file, the home's use from Sigenenergy, and the saving and payback from that use.
- The tester's file without generation data scores the bills and the best plan; use and payback need generation.
- They stay out of git: half-hourly readings show when a home is in, out and asleep. To put one in the golden test, its owner has to agree to the anonymised file being committed. The tester has to agree for theirs.
- `add_home.py --selftest` builds the tester's case with five-minute Sigenenergy-style rows stamped at the end, recovers the stored truth exactly (6,500 kWh, €331.50, payback 5.72), and scores through the app as B3-friend does.

How it was run

Each build: `SCENARIO_OUT=<dir> npx playwright test meter-scenarios --grep @scenarios` (all 71, the plan list and date frozen), then `python3 tests/fixtures/meter-scenarios/score.py <dir>` and `make_golden.py <dir>`. The scores and golden.json after each fix are in `stages/`. The explanation runs are in `explain/`: the app's own figures were read out of the running page (`app/`, from the build before the fixes, with the probe specs kept as .txt), and the truth's were rerun with one habit at a time (`shift_test.py`, `truth_parts.py`, `b1_parts.py`).